# Activity 12 - Register and Canary a Model Version

Course: Google Professional Machine Learning Engineer Training Course code: C740 Version: v2.0 | 12 September 2026 Evidence: activity artifacts

## Objective

Create registry metadata, traffic split, guardrails and rollback evidence.

## Deliverables

- model_manifest.json
- deployment.yaml
- canary_metrics.csv
- evaluate_canary.py
- A completed evidence/ folder with decision notes and verification output

## Prerequisites

- Python 3.11 or later and a terminal; Docker only where the activity explicitly uses it.
- For optional live Google Cloud steps: the safe training project created in Activity 01.
- Start in this activity folder so relative paths and cleanup commands remain scoped.

## Safety, cost and privacy

- Use a training project that contains no production or personal data.
- Replace every placeholder such as <PROJECT_ID> and <REGION> before a cloud command.
- Do not paste API keys or service-account JSON into notebooks, scripts or screenshots.
- Prefer Cloud Shell and Application Default Credentials. Never commit .env or credentials.
- Verify the result against the acceptance checklist, then run the cleanup section.

## Evidence to retain

Create an evidence/ folder locally. Save command output, job or revision IDs, metric tables, screenshots without secrets, and a short decision note. Use filenames that include the activity number and date. Evidence is part of the learning outcome, not an optional extra.

## Detailed step-by-step guide

### Step 1 - Inspect model_manifest.json; verify model version, code SHA, data version, metrics and owner are complete

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```
python3 -m json.tool model_manifest.json
```

Check: Record the observable result or explain the error before continuing.

### Step 2 - Review deployment.yaml; locate the champion/candidate traffic split and rollback guardrails

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Check: Record the observable result or explain the error before continuing.

### Step 3 - Run the canary evaluator against canary_metrics.csv

Run the canary evaluator against canary_metrics.csv.

```
python3 evaluate_canary.py --metrics canary_metrics.csv --config deployment.yaml
```

Check: Record the observable result or explain the error before continuing.

### Step 4 - Explain whether the candidate should be promoted, held or rolled back using per-version metrics

Explain whether the candidate should be promoted, held or rolled back using per-version metrics.

Check: Record the observable result or explain the error before continuing.

### Step 5 - Change traffic to 25% only after the 10% stage passes; never overwrite prior stage evidence

Change traffic to 25% only after the 10% stage passes; never overwrite prior stage evidence.

Check: Record the observable result or explain the error before continuing.

### Step 6 - Simulate a latency breach and confirm the evaluator returns a rollback decision

Simulate a latency breach and confirm the evaluator returns a rollback decision.

```
python3 evaluate_canary.py --metrics canary_metrics.csv --config deployment.yaml --simulate-
breach
```

Check: Record the observable result or explain the error before continuing.

### Step 7 - Record registry ID, immutable artifact checksum, traffic split, decision and rollback test

Record registry ID, immutable artifact checksum, traffic split, decision and rollback test.

Check: Record the observable result or explain the error before continuing.

## Acceptance checklist

[ ] Every required artifact exists and opens without error.

[ ] Results are tied to the supplied mock data or an approved training project.

[ ] The conclusion cites metrics, IDs or command output rather than opinion alone.

[ ] No credentials, tokens, personal data or unredacted sensitive prompts are present.

[ ] Cleanup is completed or the retained resource is documented with an owner and expiry.

## Troubleshooting

| Symptom | Likely cause | Corrective action |
|---|---|---|
| 403 PERMISSION_DENIED | API disabled, wrong project, or IAM role missing | Verify gcloud config get-value project, API state and the least-privilege role. |
| 429 RESOURCE_EXHAUSTED | Quota or rate limit reached | Stop retries, inspect quota, add exponential backoff, and ask the trainer before increasing quota. |
| Region/location mismatch | Dataset, bucket, model or endpoint uses another location | Recreate only the training resource in one approved region; do not copy sensitive data casually. |
| Result differs from expected | Data, dependency or configuration version drifted | Compare file hashes, pinned versions and parameters; retain the diff as evidence. |

## Cleanup

Stop local processes and remove only resources created for this activity. For live cloud work, use the Console resource views and activity evidence to verify deletion. Do not delete a shared project, dataset, bucket, registry or endpoint unless the trainer identifies it as disposable.